

WAVe Training Hyper-Parameters and Configurations

All hyperparameters remained consistent across both language experiments to ensure fair comparison and reproducibility. The only language-specific variations were in the dataset sizes, reflecting the natural differences in languages data availability.

Model Architecture

Pre-trained Encoders

- **Text Encoder:** sentence-transformers/paraphrase-multilingual-mpnet-base-v2
- **Audio Encoder:** facebook/w2v-bert-2.0

Projection Layers

- **Projection Dimension:** 768 (shared embedding space)
- **Hidden Dimension:** 1536 ($2 \times$ projection dimension)
- **Activation Function:** GELU
- **Dropout Rate:** 0.1
- **Normalization:** LayerNorm applied to final projection output

Word-Level Alignment Module

- **Text-to-Audio Attention Heads:** 6
- **Dropout Rate:** 0.1
- **Multi-Head GLU Scorer:** 4 parallel heads

Training Configuration

Core Training Parameters

- **Batch Size:** 48
- **Gradient Accumulation Steps:** 15
- **Effective Batch Size:** 720 (48×15)
- **Number of Epochs:** 30
- **Learning Rate:** $8e-4$ (main parameters)
- **Encoder Learning Rate:** $4e-5$ (discriminative rate = $\text{main_lr} / 20$)
- **Weight Decay:** 0.01
- **Random Seed:** 42

Optimization Strategy

- **Optimizer:** AdamW
- **Learning Rate Scheduler:** Linear decay with warmup
- **Warmup Steps:** 1000
- **Gradient Clipping:** Max norm = 1.0

- **Mixed Precision Training:** Disabled (Full Precision)
- **Gradient Accumulation:** Enabled with proper scheduler timing

Encoder Freezing Strategy

- **Freezing Method:** Partial unfreezing
- **Text Layers Unfrozen:** 3
- **Audio Layers Unfrozen:** 3

Loss Function Configuration

AlignmentAwareInfoNCE Loss

- **Temperature Parameter:** 0.1
- **Sentence-Level Component:** Standard InfoNCE loss
- **Word-Level Component:** Binary Cross-Entropy with soft labels
- **Uncertainty Weighting:** Learned $\log(\sigma^2)$ parameter (initialized to 0)
- **Positive Sample Labels:** 0.9 (clean transcripts)
- **Negative Sample Labels:** Variable correctness scores (0.0-0.9)

Data Processing Configuration

Audio Processing

- **Sampling Rate:** 16,000 Hz
- **Maximum Audio Length:** 480,000 samples (30 seconds)

Text Processing

- **Maximum Text Length:** 128 tokens
- **Text Padding:** Static padding to maximum length on the right side.
- **Truncation:** Enabled for sequences exceeding maximum length

Language-Specific Configurations

Dutch Experiments

- **Dataset:** Common Voice 17 Dutch (nl)
- **Synthetic Data:** 34,898 sentences generated
- **Training Set Size:** ~35K utterances

Portuguese Experiments

- **Dataset:** Common Voice 17 Portuguese (pt)
- **Synthetic Data:** 21,968 sentences generated
- **Training Set Size:** ~22K utterances

